MATH MA'AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 4: MEASUREMENTS

Learning objectives: Measurement • Standard Units • Metric Conversions and Decimal System • Time • Temperature • Money

4.1 MEASUREMENT

Measurement is the process of finding the size or quantity of an object or event using a standard unit. We measure length, weight (mass), capacity, time, temperature and money.

• **Length:** tells how long, tall or far something is.

• **Weight / mass:** tells how heavy an object is.

• **Capacity:** tells how much liquid a container can hold.

• **Time:** tells the duration or moment of an event.

• **Temperature:** tells how hot or cold something is.

4.2 STANDARD UNITS OF MEASUREMENT

The metric system is based on units that change by powers of 10. The standard (base) units used in this chapter are metre, gram and litre.

Measurement	Larger units	Base unit	Smaller units
Length	km, hm, dam	metre (m)	dm, cm, mm
Weight / mass	kg, hg, dag	gram (g)	dg, cg, mg
Capacity	kL, hL, daL	litre (L)	dL, cL, mL

Metric prefixes: kilo = 1000 base units; hecto = 100; deca = 10; deci = 1/10; centi = 1/100; milli = 1/1000.

Conversion rule: Moving one step towards a smaller unit → multiply by 10. Moving one step towards a larger unit → divide by 10. Count the number of steps before converting.

Important standard conversions

Length	Weight / mass	Capacity
1 km = 1000 m	1 kg = 1000 g	1 kL = 1000 L
1 m = 100 cm	1 g = 1000 mg	1 L = 1000 mL
1 cm = 10 mm	1 g = 10 dg	1 L = 10 dL
1 m = 1000 mm	1 kg = 10 hg	1 L = 100 cL

Other units mentioned in the textbook

• Length: 1 inch = 2.54 cm; 1 foot = 12 inches; 1 yard = 36 inches.

• Weight: 1 ounce ≈ 28 g; 1 pound = 16 ounces; 1 ton = 2000 pounds.

4.3 MEASURING UNITS AND DECIMAL SYSTEM

A measurement can be written using a decimal when a smaller unit is converted into a larger unit. For example, 854 m = 0.854 km because 1000 m = 1 km.

Examples of conversion

Given measurement	Conversion	Answer
8 km 854 m	8 + 854/1000 km	8.854 km
9 kg 58 g	9 + 58/1000 kg	9.058 kg
11 L 585 mL	11 + 585/1000 L	11.585 L
5 m 7 cm	5 + 7/100 m	5.07 m

Addition and subtraction of measurements

• Convert measurements into the same unit before adding or subtracting, OR arrange the same units in columns.

• For decimal measurements, line up the decimal points before calculating.

• Regroup when necessary, just as in ordinary addition and subtraction.

Example – Addition: 18 m 50 cm + 15 m 20 cm = 33 m 70 cm.
Example – Subtraction: 5 kg 250 g – 2 kg 875 g = 2 kg 375 g.

Word-problem method

For a question involving several quantities, first convert all quantities to a common unit. Perform the required operation and convert the final answer into the requested unit.

Example: A tank has 15,000 L. 800 students each use 5 L in a day. Water used = 800 × 5 = 4,000 L. Water left = 15,000 – 4,000 = **11,000 L**.

4.4 TIME

Time is measured using seconds, minutes, hours, days, weeks, months and years.

Unit	Equivalent
1 minute	60 seconds
1 hour	60 minutes = 3600 seconds
1 day	24 hours
1 week	7 days
1 year	12 months = 365 days (366 in a leap year)

Converting time units

• Days to hours: multiply by 24. Example: 8 days = 8 × 24 = 192 hours.

• Hours to days: divide by 24. Example: 48 hours = 48 ÷ 24 = 2 days.

• Hours to minutes: multiply by 60. Example: 8 hours = 480 minutes.

• Minutes to hours: divide by 60. Example: 240 minutes = 4 hours.

• Minutes to seconds: multiply by 60. Seconds to minutes: divide by 60.

12-hour and 24-hour clock

• **a.m.** is used from midnight up to, but not including, noon. **p.m.** is used from noon up to, but not including, midnight.

• 12-hour to 24-hour: for p.m. times after 12 noon, add 12 to the hour. Example: 5:30 p.m. = 17:30.

• For a.m. times, keep the hour; use two digits when needed. Example: 5:00 a.m. = 05:00.

• 24-hour to 12-hour: 00:00 is midnight; 12:00 is noon. For times greater than 12:00, subtract 12 and use p.m. Example: 18:40 = 6:40 p.m.

Elapsed time

Elapsed time = finishing time – starting time. If an event crosses midnight, calculate the time up to midnight and then add the time after midnight.

Example: From 22:35 to 24:00 = 1 h 25 min; from 00:00 to 01:20 = 1 h 20 min. Total duration = **2 h 45 min**.

Remember: 1 minute = 60 seconds; 1 hour = 60 × 60 = 3600 seconds. Do not treat 1 hour as 100 minutes.

4.5 TEMPERATURE

Temperature is the measure of the warmth or coldness of an object or surroundings. A **thermometer** is used to measure temperature.

• Temperature is measured in degrees.

• **Degree Celsius (°C)** and **degree Fahrenheit (°F)** are two commonly used temperature scales.

• Read the thermometer at the level of the top of the liquid column and note the scale markings.

• An increase in temperature is found by subtraction: final temperature – initial temperature.

• A decrease in temperature is found by subtraction: initial temperature – final temperature.

Example: Morning temperature = 18°C; afternoon temperature = 32°C. Increase = 32 – 18 = **14°C**.

At this level: Focus on reading thermometer scales and calculating temperature increases or decreases. Celsius and Fahrenheit are different scales; do not treat their numerical readings as interchangeable.

4.6 MONEY

In India, money is commonly expressed in rupees (₹) and paise (p).

Basic conversion: ₹1 = 100 paise. Rupees to paise → multiply by 100. Paise to rupees → divide by 100.

Examples:
₹18.75 = 18.75 × 100 = **1875 paise**.
8750 paise = 8750 ÷ 100 = **₹87.50**.

Addition and subtraction of money

• Write the amounts one below another, keeping decimal points aligned.

• Add or subtract rupees and paise in their correct place values.

• Write the final answer with the rupee symbol.

Example: ₹125.75 + ₹86.50 = **₹212.25**.

Unitary method

The unitary method finds the value of one item first, then uses it to find the value of the required number of items.

Steps:
1. Find the cost of 1 item by dividing the total cost by the number of items.
2. Multiply the cost of 1 item by the required number of items.

Example: 8 caps cost ■120.
Cost of 1 cap = ■120 ÷ 8 = ■15.
Cost of 15 caps = ■15 × 15 = ■**225**.

Key idea: Division finds the value of one item. Multiplication finds the value of many items at the same rate.

QUICK REVISION – IMPORTANT FORMULAE

Topic	Rule / Formula
Metric conversion to smaller unit	Multiply by 10 for each step
Metric conversion to larger unit	Divide by 10 for each step
Days and hours	Days × 24 = hours
Hours and minutes	Hours × 60 = minutes
Minutes and seconds	Minutes × 60 = seconds
Elapsed time	Finishing time – Starting time
Rupees to paise	Rupees × 100 = paise
Paise to rupees	Paise ÷ 100 = rupees
Unitary method	Value of one item × required quantity

IMO preparation checklist: Practise mixed-unit conversions, decimal forms of measurements, addition and subtraction with regrouping, multi-step word problems, elapsed time across midnight, 12-hour/24-hour clock conversions, thermometer readings, money calculations and unitary-method questions.

ALL THE BEST